

Varnitha Kurli

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SUMMARY

Quantitative social scientist and demographer with 9+ years in survey research, causal inference, and large-scale geospatial data analysis. Owns research end-to-end, from survey design and data collection to modeling and executive communication. Experienced in aligning government, NGO, and academic partners, with a research focus on how climate and policy shocks shape fertility, migration, mortality, and child health.

SKILLS

Methods: Survey design and analysis (cross-sectional, pooled multi-round, phone surveys), causal inference (difference-in-differences, matching), regression and quantile modeling (GLMs, GAMs), demographic methods, geospatial analysis
Tools: R, Python, Stata, SQL, Google Earth Engine, ArcGIS, MATLAB **Languages:** Telugu, Hindi, Urdu

EDUCATION

Ph.D., Environmental Science (Climate and Demographic Methods), CU Boulder	<i>Exp. May 2027</i>
M.Sc., Logic, Computation, and Methodology , Carnegie Mellon University	<i>2017</i>
B.A. (Honors), Economics and Mathematics , University of Massachusetts Amherst	<i>2015</i>

EXPERIENCE

Data Scientist Consultant | The World Bank *Aug 2022 – Present*

- Building multi-hazard geospatial risk assessments for critical infrastructure across sectors and countries using the Infrascan framework, producing exposure, vulnerability, and risk outputs for task teams (GFDRR).
- Scaled a global [gender-equity data platform](#) on water-utility employment from 25 to 196 institutions in 50+ countries, running the survey pipeline with government focal points across all 6 World Bank regions.
- Leading a gender and social-inclusion tagging database of World Bank water projects; translate findings into executive presentations for senior stakeholders.
- Core data team member for the [Gaza Interim Rapid Damage and Needs Assessment](#) (World Bank, EU, UN), coordinating cross-sector damage and loss analysis.
- Co-led a multi-track analytics program for India's Green, Resilient, and Inclusive Cities initiative, aligning ML-driven deliverables with the National Institute of Urban Affairs, city governments, and NGOs.
- Co-led chapter delivery for the [Western Balkans Climate Change and Development Report](#), coordinating European Space Agency engineers, adaptation researchers, and regional foundations.

Graduate Researcher, PhD Dissertation | University of Colorado Boulder *Aug 2022 – Present*

- **Policy and fertility:** Evaluated India's two-child policies across three adoption waves using matched difference-in-differences on pooled NFHS survey rounds, testing moderation by son preference and poverty. *Under review, Population and Development Review.*
- **Climate and displacement:** Mapped footprints for 3,500+ disasters globally (2014–2023) to estimate displacement and mortality rates per exposed population with quantile GAMs, testing conflict and development moderators. *Preparing for submission to Nature Climate Change.*
- **Compound hazards and child health:** Linking geocoded Bangladesh DHS (2017–18, 2022) child histories to satellite-derived flood, cyclone, drought, and heat exposure to test whether sequential hazards compound effects on wasting and under-5 mortality.
- **Mortenson Center in Global Engineering & Resilience:** Translating groundwater analytics into decision tools for water-dependent communities in Western Colorado.
- **CU Population Center:** Coordinating research on South Asian population policy with NGO and academic partners.

Water Data Specialist | The World Bank *Jul 2021 – Jul 2022*

- Designed and ran a multi-region farmer phone survey in Myanmar ([Food Security Brief](#)) measuring food-security impacts of climate shocks, COVID-19, and political disruption.
- Delivered interactive data applications and open repositories ([World Bank Water Data](#)) for the Global Water and Sanitation Partnership; presented water-security findings in bilateral government meetings and workshops.

- Led production of the Cities and Climate Change Report, synthesizing cross-sector climate risk mechanisms into risk-management policy recommendations.

Data Analyst | The World Bank

Dec 2018 – Jun 2021

- Built near real-time economic indicators from Twitter and location data to track COVID-19 recovery, including [Multilingual Detection of Employment Status](#).
- Supported spatial analysis and logistics for the West Bengal Inland Water Transport and Logistics project.

Research Assistant | Bharti Institute of Public Policy, ISB

Jun 2017 – Jun 2018

- Managed data collection and analysis for NSF- and NASA-funded projects on environmental management and afforestation in India.

PUBLICATIONS

Published

- Fischer, H.W., et al. (incl. **Kurli, V.**) (2025). Forest restoration for the environment and human well-being is supported by empowered local governance over long time horizons. *Environmental Research Letters*.
- Demaree, K., **Kurli, V.**, Magnuszewski, P., Andersson, K., & Thomas, E. (2024). Development and evaluation of a digital behavioral economics game towards improved understanding of groundwater conservation in Southern Colorado. *PLOS Water*, 3(12), e0000298.
- Kaur, K., **Kurli, V.**, & Andersson, K. (2025). The challenge of governing complex forest ecosystems: Can a polycentric approach help? In *Handbook of Institutions and Complexity*. Edward Elgar.

Under review

- **Kurli, V.**, & Menken, J. State-level two-child policies and progression to a third child in India: Evidence from three policy waves. *Population and Development Review*.
- Braich, G., . . . , **Kurli, V.**, & Mehrabi, Z. Two-thirds of new fossil fuel infrastructure targets critical ecosystems, elevating health risks for millions. *Nature*.
- Agrawal, A., Chhatre, A., Gerber, E., & **Kurli, V.** Behavioral and motivational crowding undermine sustainable development programs. *Nature Human Behaviour*.

In preparation

- **Kurli, V.**, Carrico, A., & Mehrabi, Z. Mapping climate-driven internal displacement and moderating factors globally.
- **Kurli, V.**, & Carrico, A. Compounding climate health shocks and child health in Bangladesh.
- Menken, J., **Kurli, V.**, Khan, M.N., & Williams, J. Changing expression of gender preference: The case of Bangladesh.

HONORS & FELLOWSHIPS

CU Boulder Chancellor's Fellowship (\$61K) • CU Population Center Seed Grant (\$25K) • ENVIS Summer Research Fellowship (\$18K) • Co-Authorship Grant (\$2K) • IBS Poster Symposium 1st Prize (2023, 2024) • LSEC Fellowship, Carnegie Mellon (2016) • Economics Alumni Award for Outstanding Achievement, UMass Amherst